Randomized Algorithms

The Probabilistic Method (I)

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Outline

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Motivation

- Prove the existence of objects.

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- Prove the existence of objects.
- If the probability of selecting an object with the required properties is positive, then the sample space must contain such an object.

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Example

- Coloring the edges of a graph with two **colors**.

  Constraint: no large cliques with all edges having the same color.

Theorem 1

If $\binom{n}{k}2^{-\binom{k}{2}+1} < 1$, then it is possible to color the edges of K_n with two colors so that it has no monochromatic K_k subgraph.

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  - There are $\binom{n}{k}$ different K_k cliques of K_n.

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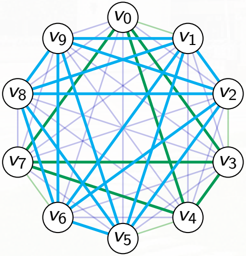

- $G_{\{v_0,v_3,v_4,v_7\}}$: a monochromatic K_4.
- $G_{\{v_1,v_2,v_5,v_6,v_8,v_9\}}$: a monochromatic K_6.
- No monochromatic K_k for $k \geq 7$.

General Idea

- Let A_i be the event that clique i is monochromatic.
- **Goal:** Prove that the probability is $\Pr\left[\bigcap_{i=1}^{\binom{n}{k}} \overline{A_i}\right] > 0$.

General Idea

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- **Goal:** Prove that the probability is $\Pr\left[\bigcap_{i=1}^{\binom{n}{k}} \overline{A_i}\right] > 0$.
  - That is,

$$1 - \Pr\left[\bigcup_{i=1}^{\binom{n}{k}} A_i\right] > 0.$$

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- By the union bound,

$$\Pr\left[\bigcup_{i=1}^{\binom{n}{k}} A_i\right] \leq \sum_{I=1}^{\binom{n}{k}} \Pr[A_i] = \binom{n}{k} 2^{-\binom{k}{2}+1} < 1,$$

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- Hence

$$\Pr\left[\bigcap_{i=1}^{\binom{n}{k}} \overline{A_i}\right] = 1 - \Pr\left[\bigcup_{i=1}^{\binom{n}{k}} A_i\right] > 0.$$

For sufficiently large k

- Our calculations can be simplified if we note that:
  For $n \leq 2^{k/2}$ and $k \geq 3$,

$$\begin{aligned}\binom{n}{k} 2^{-\binom{k}{2}+1} &\leq \frac{n^k}{k!} 2^{-(k(k-1)/2)+1} \\ &\leq \frac{2^{k/2+1}}{k!} \\ &< 1.\end{aligned}$$

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- **Example:** Finding a coloring on a graph of 1,000 vertices with no monochromatic K_{20}.
  - The probability that a random coloring has a monochromatic K_{20} is

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$\Rightarrow$ A Monte Carlo algorithm with a very small probability of failure.

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  - A polynomial time verifier is required.
- Expected number of samples: $1/p$.
- For a fixed (constant) k, check all $\binom{n}{k}$ cliques and make sure they are not monochromatic.
  - Not polynomial time when k grows with n!
  - $O(n^k k^2) = O^*(n^{f(k)})$ time: **XP** (eXtendable Polynomial time).

Outline

- To prove that an object with certain properties exists, we can design a probability space from which an element chosen at random yields an object with the desired properties with a positive probability.
- A similar and sometimes easier approach: Using an averaging argument.

The expectation (average) argument

- **Intuitive idea:** Say a discrete random variable X has $\mathbb{E}[X] = \mu$. Then there must be some value $a \leq \mu$ and $b \geq \mu$ such that $\Pr[X = a] > 0$ and $\Pr[X = b] > 0$.

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Lemma 1

Suppose we have a probability space S such and a random variable X defined on S such that $\mathbb{E}[X] = \mu$. Then

$$\Pr[X \geq \mu] > 0 \quad \text{and} \quad \Pr[X \leq \mu] > 0.$$

Proof: Let $\mathcal{X} = \{x \in \mathbb{R} : \Pr[X = x] > 0\}$ be the support of X. Suppose that

$$\mu = \mathbb{E}[X] = \sum_{x \in \mathcal{X}} \Pr[X = x].$$

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Similarly, if $\Pr[X \leq \mu] = 0$ then

$$\mu = \sum_{x \in \mathcal{X}} x \Pr[X = x] = \sum_{x > \mu} x \Pr[X = x] > \sum_{x > \mu} \mu \Pr[X = x] = \mu.$$

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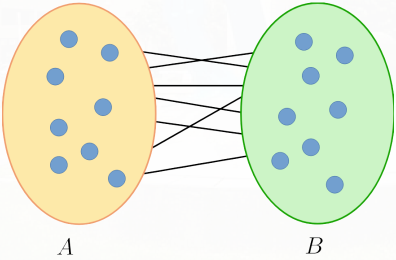
Application 1: Finding a Large Cut

Theorem 2

Given an undirected graph $G = (V, E)$ with $|V| = n$ and $|E| = m$. There is a partition of V into disjoint sets A and B such that at least $m/2$ edges connect a vertex in A to a vertex in B. That is, there is a cut with value at least $m/2$.



Proof

- Construct sets A and B by randomly and independently assigning each vertex to one of the two sets.
- Let $e_1, e_2, \ldots, e_m$ be an arbitrary enumeration of the edges of G.
- For $i = 1, \ldots, m$, define X_i such that

$$X_i = \begin{cases} 1 & \text{if edge } i \text{ connects } A \text{ to } B \\ 0 & \text{otherwise} \end{cases}$$

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- The probability that edge e_i connects a vertex in A to a vertex in B is $\frac{1}{2}$, and thus

$$\mathbb{E}[X_i] = \frac{1}{2}$$

Proof (contd.)

- Let $C = (A, B)$ be a random variable denoting the value of the cut corresponding to the sets A and B. Then

$$\mathbb{E}[C(A,B)] = \mathbb{E}\left[\sum_{i=1}^{m} X_i\right] = \sum_{i=1}^{m} \mathbb{E}[X_i] = m \cdot \frac{1}{2} = \frac{m}{2}.$$

How to efficiently find a cut with value $\geq m/2$?

- We need "a lower bound" on the probability that a random partition has a cut of value at least $m/2$.
- To derive such a bound, let $p = \Pr\left[C(A,B) \geq \frac{m}{2}\right]$, and observe that $C(A,B) \leq m$, Then

$$\begin{aligned}
\frac{m}{2} &= \mathbb{E}[C(A,B)] \\
&= \sum_{i<m/2} i \Pr[C(A,B)=i] + \sum_{i \geq m/2} i \Pr[C(A,B)=i] \\
&\leq (1-p)\left(\frac{m}{2}-1\right) + pm,
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which implies that $p \geq \frac{1}{m/2+1}$ $\Rightarrow$ expected number of samples before finding a cut: $\leq m/2+1$.

Application 2: Maximum Satisfiability

The following expression is an instance of SAT:

$$(x_1 \vee \overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_3}) \wedge (x_1 \vee x_2 \vee x_4) \wedge (x_4 \vee \overline{x_3}) \wedge (x_4 \vee \overline{x_1}).$$

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Theorem 3

Given a set of m clauses, let k_i be the number of literals in the ith clause for $i = 1, 2, \ldots, m$. Let $k = \min_{i \in \{1,2,\ldots,m\}} k_i$. Then there is **a truth assignment that satisfies at least**

$$\boldsymbol{\sum_{i=1}^{m}(1 - 2^{-k_i}) \geq m(1 - 2^{-k})} \quad \textbf{clauses.}$$

Proof

- Assign values independently and uniformly at random to the variables.
- The probability that the ith clause with k_i literals is satisfied is at least $(1 - 2^{-k})$.

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- The expected number of satisfied clauses is therefore at least

$$\sum_{i=1}^{m}(1 - 2^{-k_i}) \geq m(1 - 2^{-k}),$$

and there must be an assignment that satisfies at least that many clauses.

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- The probabilistic method can yield insight into how to construct **deterministic** algorithms.
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  - x_i: the set where v_i is placed (so x_i is either A or B).
  - **Assumption:** The first k vertices have been placed. Consider the expected cut value if the *remaining* vertices are then placed *independently and uniformly into one of the two sets*.

Derandomization by Conditional Expectation (1/4)

- We show inductively how to place the next vertex so that

$$\mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k] \leq \mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_{k+1}]$$

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- **The right-hand side:** the value of the cut determined by our placement algorithm.
- Hence our algorithm returns a cut whose value $\geq \mathbb{E}[C(A,B)]) \geq m/2$.

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- Consider placing v_{k+1} randomly, so that it is placed in A or B with probability $1/2$ each.
- Let Y_{k+1} be a random variable representing the set where it is placed.

Derandomization by Conditional Expectation (3/4)

- Then

$$
\begin{aligned}
\mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k] = {} & \frac{1}{2}\mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k, Y_{k+1} = A] \\
& + \frac{1}{2}\mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k, Y_{k+1} = B].
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- It follows that

$$\max\Big(\mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k, Y_{k+1} = A],\ \mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k, Y_{k+1} =$$

$$\geq \mathbb{E}[C(A,B) \mid x_1, x_2, \ldots, x_k].$$

Derandomization by Conditional Expectation (4/4)

- Compute the two quantities
  - $\mathbb{E}[C(A,B) \mid x_1, \ldots, x_k, Y_{k+1} = A]$
  - $\mathbb{E}[C(A,B) \mid x_1, \ldots, x_k, Y_{k+1} = B]$

  and then place v_{k+1} in the set that yields the larger expectation.
- Once we do this, we will have a placement satisfying

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Computation of $\mathbb{E}[C(A,B) \mid x_1, \ldots, x_k, Y_{k+1} = A \text{ (or } B)]$

- For the first $k+1$ vertices, compute the number of edges that contribute to the cut.
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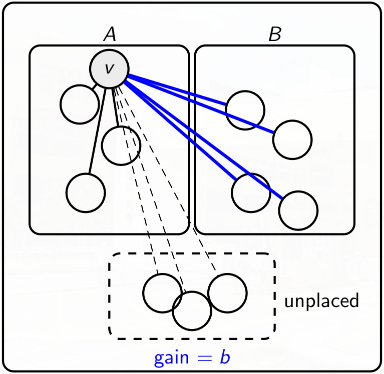
- For the first $k+1$ vertices, compute the number of edges that contribute to the cut.
- For all the other edges, each one contributes to the cut with prob. $\frac{1}{2}$.
  - Depends on whether v_{k+1} has more neighbors in A or in B.

Simple greedy derandomized algorithm

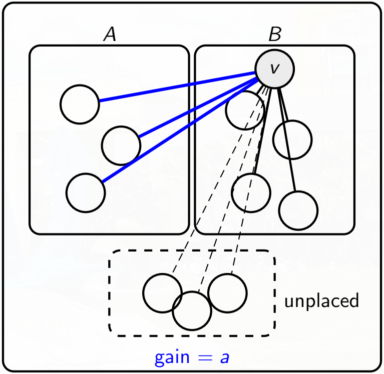
1. Take the vertices in some order.
2. Place the first vertex arbitrarily in A.
3. Place each successive vertex to maximize the number of edges crossing the cut.
    - Equivalently, place each vertex on the side with fewer neighbors, breaking ties arbitrarily.

Two choices for v_{k+1}

Place v_{k+1} in A



Place v_{k+1} in B



$$\mathbb{E}[C \mid x_1, x_2, \ldots, x_k, v_{k+1} = A] = b, \quad \mathbb{E}[C \mid x_1, x_2. \ldots, x_k, v_{k+1} = B] = a.$$

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Sample and Modify

Sample and Modify (Two Stages)

- 1st Stage: Construct a random structure that does not have the required properties.
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- In some cases, it is easy to work using this indirect approach.

Application 1: Independent Sets

Theorem 4

Let $G = (V, E)$ be a graph on n vertices with m edges. Then G has an independent set with at least $n^2/4m$ vertices.

Proof: Let $d = 2m/n$ be the average degree of the vertices in G. Consider the following randomized algorithm.

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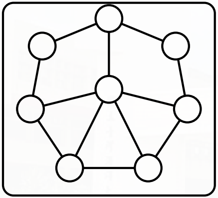
2. For each remaining edge, remove it and one of its adjacent vertices.

The remaining vertices form an independent set (all edges have been removed).

Sample & Modify for Independent Sets

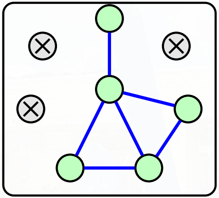
1. Original graph G



Delete each vertex independently

with survival probability $1/d$.

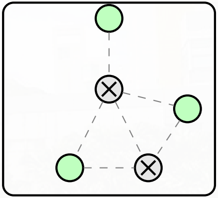
2. After random sampling



Let X be the number of surviving vertices,

and let Y be the number of surviving edges.

3. After modification



Remove one endpoint from each surviving edge.

The remaining green vertices form an independent set.

$$\mathbb{E}[X] = \frac{n}{d}, \qquad \mathbb{E}[Y] = \frac{n}{2d}, \qquad \Rightarrow \qquad \mathbb{E}[X - Y] = \frac{n}{2d} = \frac{n^2}{4m}.$$

Proof (2/3)

- Let X be the number of vertices that survive the first step of the algorithm.
- Since the graph has n vertices and each vertex survives with probability $1/d$, it follows that

$$\mathbb{E}[X] = \frac{n}{d}.$$

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- Let Y be the number of edges that survive the first step. There are $nd/2$ edges in the graph, and an edge survives iff its two adjacent vertices survive.
- Thus,

$$\mathbb{E}[Y] = \frac{nd}{2}\left(\frac{1}{d}\right)^2 = \frac{n}{2d}.$$

Proof (3/3)

- The second step of the algorithm removes all the remaining edges and at most Y vertices.
- When the algorithm terminates, it outputs an independent set of size at least $X - Y$, and

$$\mathbb{E}[X - Y] = \frac{n}{d} - \frac{n}{2d} = \frac{n}{2d}.$$

- The expected size of the independent set generated by the algorithm is $n/2d$, so the graph has an independent set with at least $n/2d = n^2/4m$ vertices.

Application 2: Graphs with Large Girth

Theorem 5

For any integer $k \geq 3$, there is a graph with n nodes, at least $\frac{1}{4}n^{1+1/k}$ edges, and girth at least k.

- girth: the length of a **shortest cycle** contained in the graph.

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- girth: the length of a **shortest cycle** contained in the graph.

Proof: We first sample a random graph $G \in G_{n,p}$ with $p = n^{1/k-1}$. Let X be the number of edges in the graph. Then

$$\mathbb{E}[X] = p\binom{n}{2} = \frac{1}{2}\left(1 - \frac{1}{n}\right)n^{1/k+1}.$$

- $G_{n,p}$: a set of graphs of n vertices where every (u, v) exists with prob. p.

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  - First, consider choosing the i vertices, then consider the possible orders, and finally keep in mind that reversing the order yields the same cycle.

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- Also, there are $\binom{n}{i}\frac{(i-1)!}{2}$ possible cycles of length i.
  - First, consider choosing the i vertices, then consider the possible orders, and finally keep in mind that reversing the order yields the same cycle.
- Hence, (recall that $p = n^{1/k-1}$)

$$\begin{aligned}\mathbb{E}[Y] = \sum_{i=3}^{k-1}\binom{n}{i}\frac{(i-1)!}{2}p^i \quad &\leq \quad \sum_{i=3}^{k-1} n^i p^i = \sum_{i=3}^{k-1} n^i (n^{1/k-1})^i \\ &= \quad \sum_{i=3}^{k-1} n^{i/k} < kn^{(k-1)/k}.\end{aligned}$$

Proof (3/3)

X: # edges in the graph; Y: # cycles in the graph of length $\leq k-1$.

- We modify the original randomly chosen graph G by eliminating one edge from each cycle of length $\leq k-1$.
- The modified graph has girth at least k ($\because$ shorter cycles are destroyed).

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- We modify the original randomly chosen graph G by eliminating one edge from each cycle of length $\leq k-1$.
- The modified graph has girth at least k ($\because$ shorter cycles are destroyed).
- When n is sufficiently large, the expected number of edges in the resulting graph is

$$\mathbb{E}[X-Y] \geq \frac{1}{2}\left(1-\frac{1}{n}\right)n^{1/k+1} - kn^{(k-1)/k} \geq \frac{1}{4}n^{1/k+1}.$$

- So, there exists a graph with $\geq \frac{1}{4}n^{1+1/k}$ edges and girth $\geq k$.

Outline

The Second Moment Method

- In the $G_{n,p}$ model, it is often the case that there is a threshold function f such that
  - when $p = O(f(n))$ or $p = o(f(n))$, **almost no** graph has the desired property;
  - when $p = \Omega(f(n))$ or $p = \omega(f(n))$, **almost every** graph has the desired property.

The Second Moment Method

- The following theorem from Chebyshev's inequality is often used.

Theorem 6 [Derived from Chebyshev's Inequality]

X is a nonnegative integer-valued random variable, then

$$\Pr[X = 0] \leq \frac{\mathrm{Var}[X]}{(\mathbb{E}[X])^2}.$$

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X is a nonnegative integer-valued random variable, then

$$\Pr[X = 0] \leq \frac{\mathrm{Var}[X]}{(\mathbb{E}[X])^2}.$$

Proof:

$$\Pr[X = 0] \leq \Pr[|X - \mathbb{E}[X]| \geq \mathbb{E}[X]] \leq \frac{\mathrm{Var}[X]}{(\mathbb{E}[X])^2}.$$

Application: Threshold Behavior in Random Graphs

Theorem 7

Consider $G_{n,p}$, suppose that $p = f(n)$.

Let A be the event that a random graph chosen from $G_{n,p}$ has a clique of ≥ 4 vertices. Then, for any $\varepsilon > 0$ and sufficiently large n,

$$\Pr[A] < \varepsilon \quad \text{if } f(n) = o(n^{-2/3}).$$

Similarly, if $f(n) = \omega(n^{-2/3})$ then, for sufficiently large n,

$$\Pr[\overline{A}] < \varepsilon.$$

Proof (1/4)

- We first consider the case in which $p = f(n)$ and $f(n) = o(n^{-2/3})$.
- $C_1, C_2, \ldots, C_{\binom{n}{4}}$: an enumeration of all the subsets of 4 vertices in G.
- Let

$$X_i = \begin{cases} 1 & \text{if } C_i \text{ is a 4-clique} \\ 0 & \text{otherwise.} \end{cases}$$

- Let

$$X = \sum_{i=1}^{\binom{n}{4}} X_i,$$

so that (A 4-clique has 6 edges)

$$\mathbb{E}[X] = \binom{n}{4} p^6$$

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so that (A 4-clique has 6 edges)

$$\mathbb{E}[X] = \binom{n}{4} p^6 = o(1).$$

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- Since X is a nonnegative integer-valued random variable, it follows that $\Pr[X \geq 1] \leq \mathbb{E}[X] < \varepsilon$.

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- So, in this case $\mathbb{E}[X] = o(1) \Rightarrow \mathbb{E}[X] < \varepsilon$ for sufficiently large n.
- Since X is a nonnegative integer-valued random variable, it follows that $\Pr[X \geq 1] \leq \mathbb{E}[X] < \varepsilon$.
- Hence, the probability that a random graph chosen from $G_{n,p}$ has a clique of ≥ 4 vertices is less than ε.

Proof (3/4)

- Next, consider the case when $p = f(n) = \omega(n^{-2/3})$.
- $\mathbb{E}[X] \to \infty$ as n grows large.
  - Not sufficient to conclude that, with high probability, a graph chosen random from $G_{n,p}$ has a clique of ≥ 4 vertices.
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- Then, we must show that $\mathrm{Var}[X] = o((\mathbb{E}[X])^2)$.
- Consider the following lemma:

Proof (4/4)

Lemma 2

Let $Y_i, i = 1, 2, \ldots, m$, be 0-1 random variables, and let $Y = \sum_{i=1}^{m} Y_i$. Then

$$\mathrm{Var}[Y] \leq \mathbb{E}[Y] + \sum_{1 \geq i,j \geq m;\ i \neq j} \mathrm{Cov}(Y_i, Y_j).$$

Proof of Lemma 2

- For any sequence of random variables $Y_1, Y_2, \ldots, Y_m$,

$$\mathrm{Var}\left[\sum_{i=1}^{m} Y_i\right] = \sum_{i=1}^{m} \mathrm{Var}[Y_i] + \sum_{1 \le i,j \le m; i \ne j} \mathrm{Cov}(Y_i, Y_j).$$

- This is the generalization from two variables to m variables.
- When Y_i is a 0-1 random variable, $\mathbb{E}[Y_i^2] = \mathbb{E}[Y_i]$ and so

$$\mathrm{Var}[Y_i] = \mathbb{E}[Y_i^2] - (\mathbb{E}[Y_i])^2 \le \mathbb{E}[Y_i],$$

giving the lemma.

Finishing the proof of Theorem 7

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Let A be the event that a random graph chosen from $G_{n,p}$ has a clique of ≥ 4 vertices. Then, for any $\varepsilon > 0$ and sufficiently large n,

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- Compute variance of the number of 4-cliques

$$\mathrm{Var}[X] = \mathrm{Var}\left(\sum_{i=1}^{\binom{n}{4}} X_i\right).$$

Using covariance

- Since X_i is an indicator,

$$\mathrm{Var}[X_i] \le \mathbb{E}[X_i] = p^6, \quad \Rightarrow \quad \sum_i \mathrm{Var}(X_i) \le \binom{n}{4} p^6.$$

- The remaining term is the total covariance. It depends on how much the two 4-cliques C_i and C_j overlap.
- Write $C_i \cap C_j$ for their intersection. We consider the cases $|C_i \cap C_j| = 0, 1, 2, 3$.

Pairs of cliques: overlap cases (1/2)

- **Case** $|C_i \cap C_j| = 0$ **or** 1**.**
  - The sets of edges involved in the two cliques are disjoint.
  - Thus X_i and X_j are independent, so $\mathbb{E}[X_iX_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] = 0$.
- **Case** $|C_i \cap C_j| = 2$**.**
  - The cliques share one edge; altogether $\binom{4}{2} - 1 = 11$ distinct edges must be present.
  - Hence $\mathbb{E}[X_iX_j] \le p^{11} \quad \Rightarrow \quad \mathbb{E}[X_iX_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] \le p^{11}$.
  - The number of ordered pairs (C_i, C_j) with $|C_i \cap C_j| = 2$ is

$$\binom{n}{6}\binom{6}{2,2,2}.$$

Choose 6 vertices and then split them into $C_i \cap C_j$ (2 vertices), 2 vertices for $C_i \setminus C_j$, and 2 for $C_j \setminus C_i$.

Pairs of cliques: overlap cases (2/2)

- **Case $|C_i \cap C_j| = 3$.**
  - The cliques share three vertices (a triangle), hence $\binom{4}{2} - 3 = 9$ distinct edges must appear.
  - Thus
  $$\mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] \le \mathbb{E}[X_i X_j] \le p^9.$$
  - There are
  $$\binom{n}{5}\binom{5}{3,1,1}$$
  ordered pairs (C_i, C_j) with $|C_i \cap C_j| = 3$.

Completing the bound on $\mathrm{Var}(X)$

- Collecting all contributions,

$$\mathrm{Var}(X) \le \binom{n}{4} p^6 + \binom{n}{6} \binom{6}{2,2,2} p^{11} + \binom{n}{5} \binom{5}{3,1,1} p^9.$$

- Using $p = f(n) = \omega(n^{-2/3})$ and $\mathbb{E}[X] = \binom{n}{4} p^6$, one checks that

$$\mathrm{Var}(X) = o(n^8 p^{12}) = o\big((\mathbb{E}[X])^2\big).$$

since $(\mathbb{E}[X])^2 = (\binom{n}{4} p^6)^2 = \Theta(n^8 p^{12})$.

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By the second moment method, this implies $\Pr[X = 0] = o(1)$,

$\Rightarrow$ with high probability G contains a copy of K_4.

Outline

- For a **sum of Bernoulli random variables**, there is an alternative to the second moment method.

Theorem 8

Let $X = \sum_{i=1}^{n} X_i$, where each X_i is a 0-1 random variable. Then

$$\Pr[X > 0] \geq \sum_{i=1}^{n} \frac{\Pr[X_i = 1]}{\mathbb{E}[X \mid X_i = 1]}.$$

- Note that X_i's need not to be independent here.

Proof

- Let $Y = 1/X$ if $X > 0$, with $Y = 0$ otherwise. Then

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$$\mathbb{E}[XY] = \sum_{x>0} x \cdot \frac{1}{x} \cdot \Pr[X = x] = \sum_{x>0} \Pr[X = x] = \Pr[X > 0].$$

- However,

$$\begin{aligned}
\mathbb{E}[XY] = \mathbb{E}\left[\sum_{i=1}^{n} X_i Y\right] &= \sum_{i=1}^{n} \mathbb{E}[X_i Y] \\
&= \sum_{i=1}^{n} \left(\mathbb{E}[X_i Y \mid X_i = 1] \Pr[X_i = 1] + \mathbb{E}[\boldsymbol{X_i} Y \mid \boldsymbol{X_i = 0}] \Pr[X_i = 0]\right) \\
&= \sum_{i=1}^{n} \mathbb{E}[Y \mid X_i = 1] \Pr[X_i = 1] = \sum_{i=1}^{n} \mathbb{E}[1/X \mid X_i = 1] \Pr[X_i = 1] \\
&\geq \sum_{i=1}^{n} \frac{\Pr[X_i = 1]}{\mathbb{E}[X \mid X_i = 1]}. \quad \text{(by Jensen's inequality; } \mathbb{E}[f(Z)] \geq f(\mathbb{E}[Z])\text{)}
\end{aligned}$$

An alternative proof of the 4-clique existence (1/3)

- Let $X = \sum_{i=1}^{\binom{n}{4}} X_i$, where X_i is 1 if the subset of four vertices C_i is a 4-clique and 0 otherwise.
- For a specific X_j, we have $\Pr[X_j = 1] = p^6$.

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- For a specific X_j, we have $\Pr[X_j = 1] = p^6$. Using the linearity of expectations, we compute

$$\mathbb{E}[X \mid X_j = 1] = \mathbb{E}\left[\sum_{i=1}^{\binom{n}{4}} X_i \,\middle|\, X_j = 1\right] = \sum_{i=1}^{\binom{n}{4}} \mathbb{E}[X_i \mid X_j = 1].$$

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- Conditioning on $X_j = 1$, we now compute $\mathbb{E}[X_i \mid X_j = 1]$ by using that, for a 0-1 random variable,

$$\mathbb{E}[X_i \mid X_j = 1] = \Pr[X_i = 1 \mid X_j = 1].$$

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- There are $\binom{n-4}{4}$ sets of vertices C_i that do not interest C_j. Each corresponding X_i is 1 with probability p^6.

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- Similarly, $X_i = 1$ with probability p^6 for the $4\binom{n-4}{3}$ sets C_i that have one vertex in common with C_j.
- For the remaining cases, we have
  - $\Pr[X_i = 1 \mid X_j = 1] = p^5$ for the $6\binom{n-4}{2}$sets C_i that have two vertices in common with C_j and
  - $\Pr[X_i = 1 \mid X_j = 1] = p^3$ for the $4\binom{n-4}{1}$ sets C_i that have three vertices in common with C_j.

An alternative proof of the 4-clique existence (3/3)

- Overall, we have

$$\begin{aligned}\mathbb{E}[X \mid X_j = 1] &= \sum_{i=1}^{\binom{n}{4}} \mathbb{E}[X_i \mid X_j = 1] \\ &= 1 + \binom{n-4}{4} p^6 + 4\binom{n-4}{3} p^6 + 6\binom{n-4}{2} p^5 \\ &+ 4\binom{n-4}{1} p^3.\end{aligned}$$

- Applying Theorem 8 yields (note: $\frac{\Pr[X_j=1]}{\mathbb{E}[X|X_j=1]}$ is the same $\forall j$)

$$\Pr[X > 0] \geq \frac{\binom{n}{4} p^6}{1 + \binom{n-4}{4} p^6 + 4\binom{n-4}{3} p^6 + 6\binom{n-4}{2} p^5 + 4\binom{n-4}{1} p^3},$$

which approaches 1 as $n \uparrow$ when $p = f(n) = \omega(n^{-2/3})$.

Discussions